Pinout of Raspberry Pi Pico

UART	I2C	SPI	PWM	GPIO	PIN#	PIN#	GPIO	ADC	PWM	SPI	I2C	UART
UART0 TX	I2C0 SDA	SPI0 RX	PWM 0A	GP0	1	40	VBUS	5V from USB connector, or 5V input				
UART0 RX	I2C0 SCL	SPI0 CSn	PWM 0B	GP1	2	39	VSYS	1.8–5.5V input				
				GND	3	38	GND					
UART0 CTS	I2C1 SDA	SPI0 SCK	PWM 1A	GP2	4	37	3V3 EN	Enable 3V3 power supply, pull low to disable				
UART0 RTS	I2C1 SCL	SPI0 TX	PWM 1B	GP3	5	36	3V3 OUT					
UART1 TX	I2C0 SDA	SPI0 RX	PWM 2A	GP4	6	35		AREF	3V3 Reference voltage for ADC			
UART1 RX	I2C0 SCL	SPI0 CSn	PWM 2B	GP5	7	34	GP28	ADC2	PWM 6A	SPI1 RX	I2C0 SDA	UART0 RX
				GND	8	33	GND	AGND	Ground for ADC			
UART1 CTS	I2C1 SDA	SPI0 SCK	PWM 3A	GP6	9	32	GP27	ADC1	PWM 5B	SPI1 TX	I2C1 SCL	UART1 RTS
UART1 RTS	I2C1 SCL	SPI0 TX	PWM 3B	GP7	10	31	GP26	ADC0	PWM 5A	SPI1 SCK	I2C1 SDA	UART1 CTS
UART1 TX	I2C0 SDA	SPI1 RX	PWM 4A	GP8	11	30	RUN	Pull low to reset the Pico				
UART1 RX	I2C0 SCL	SPI1 CSn	PWM 4B	GP9	12	29	GP22	-------------	PWM 3A	SPI0 SCK	I2C1 SDA	UART1 CTS
				GND	13	28	GND					
UART1 CTS	I2C1 SDA	SPI1 SCK	PWM 5A	GP10	14	27	GP21	-------------	PWM 2B	SPI0 CSn	I2C0 SCL	UART1 RX
UART1 RTS	I2C1 SCL	SPI1 TX	PWM 5B	GP11	15	26	GP20	-------------	PWM 2A	SPI0 RX	I2C0 SDA	UART1 TX
UART0 TX	I2C0 SDA	SPI1 RX	PWM 6A	GP12	16	25	GP19	-------------	PWM 1B	SPI0 TX	I2C1 SCL	UART0 RTS
UART0 RX	I2C0 SCL	SPI1 CSn	PWM 6B	GP13	17	24	GP18	-------------	PWM 1A	SPI0 SCK	I2C1 SDA	UART0 CTS
				GND	18	23	GND					
UART0 CTS	I2C1 SDA	SPI1 SCK	PWM 7A	GP14	19	22	GP17	-------------	PWM 0B	SPI0 CSn	I2C0 SCL	UART0 RX
UART0 RTS	I2C1 SCL	SPI1 TX	PWM 7B	GP15	20	21	GP16	-------------	PWM 0A	SPI0 RX	I2C0 SDA	UART0 TX

All GPIO pins support level and edge interrupts

* RP2350 Light Green = HSTX

2 × SPI, 2 × I2C, 2 × UART, 3 × 12-bit ADC, 16 × controllable PWM channels, 1 CPU temperature sensor

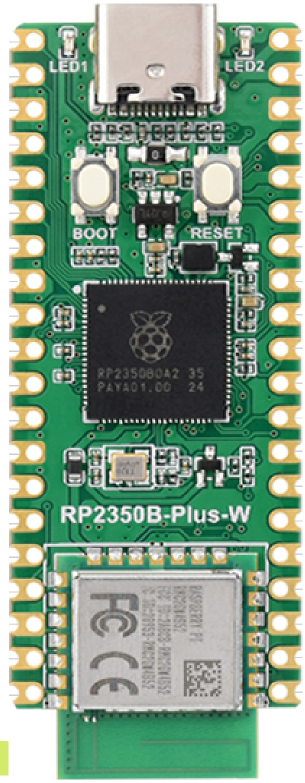
Pico: 8, Pico 2: 12 Programmable I/O (PIO) state machines

Pico 2: Floating point hardware

		GPIO	ADC	PWM	SPI	I2C	UART
ADC3 to measure VSYS		GP29	ADC3	PWM 6B	SPI1 CSn	I2C0 SCL	UART0 RX
	LED	GP25	-------------	PWM 4B	SPI1 CSn	I2C0 SCL	UART1 RX
VBUS sense - high if VBUS is present, else low		GP24	-------------	PWM 4A	SPI1 RX	I2C0 SDA	UART1 TX
3V3 Switch Mode Power Supply Power Save pin, pull low for low power mode		GP23	-------------	PWM 3B	SPI0 TX	I2C1 SCL	UART1 RTS

Pinout of Waveshare RP2350B-Plus-W

UART	I2C	SPI	PWM	GPIO	PIN#
UART0 TX	I2C0 SDA	SPI0 RX	PWM 0A	GP0	1
UART0 RX	I2C0 SCL	SPI0 CSn	PWM 0B	GP1	2
				GND	3
UART0 CTS	I2C1 SDA	SPI0 SCK	PWM 1A	GP2	4
UART0 RTS	I2C1 SCL	SPI0 TX	PWM 1B	GP3	5
UART1 TX	I2C0 SDA	SPI0 RX	PWM 2A	GP4	6
UART1 RX	I2C0 SCL	SPI0 CSn	PWM 2B	GP5	7
				GND	8
UART1 CTS	I2C1 SDA	SPI0 SCK	PWM 3A	GP6	9
UART1 RTS	I2C1 SCL	SPI0 TX	PWM 3B	GP7	10
UART1 TX	I2C0 SDA	SPI1 RX	PWM 4A	GP8	11
UART1 RX	I2C0 SCL	SPI1 CSn	PWM 4B	GP9	12
				GND	13
UART1 CTS	I2C1 SDA	SPI1 SCK	PWM 5A	GP10	14
UART1 RTS	I2C1 SCL	SPI1 TX	PWM 5B	GP11	15
UART0 TX	I2C0 SDA	SPI1 RX	PWM 6A	GP12	16
UART0 RX	I2C0 SCL	SPI1 CSn	PWM 6B	GP13	17
				GND	18
UART0 CTS	I2C1 SDA	SPI1 SCK	PWM 7A	GP14	19
UART0 RTS	I2C1 SCL	SPI1 TX	PWM 7B	GP15	20



PIN#	GPIO	ADC	PWM	SPI	I2C	UART
40	VBUS	5V from USB connector, or 5V input				
39	VSYS	1.8–5.5V input				
38	GND					
37	3V3 EN	Enable 3V3 power supply, pull low to disable				
36	3V3 OUT					
35		AREF	3V3 Reference voltage for ADC			
34	GP42	ADC2	PWM 9A	SPI1 SCK	I2C1 SDA	UART1 CTS
33	GND	AGND	Ground for ADC			
32	GP41	ADC1	PWM 8B	SPI1 CSn	I2C0 SCL	UART1 RX
31	GP40	ADC0	PWM 8A	SPI1 RX	I2C0 SDA	UART1 TX
30	RUN	Pull low to reset the Pico				
29	GP22	-------------	PWM 3A	SPI0 SCK	I2C1 SDA	UART1 CTS
28	GND					
27	GP21	-------------	PWM 2B	SPI0 CSn	I2C0 SCL	UART1 RX
26	GP20	-------------	PWM 2A	SPI0 RX	I2C0 SDA	UART1 TX
25	GP19	-------------	PWM 1B	SPI0 TX	I2C1 SCL	UART0 RTS
24	GP18	-------------	PWM 1A	SPI0 SCK	I2C1 SDA	UART0 CTS
23	GND					
22	GP17	-------------	PWM 0B	SPI0 CSn	I2C0 SCL	UART0 RX
21	GP16	-------------	PWM 0A	SPI0 RX	I2C0 SDA	UART0 TX

All GPIO pins support level and edge interrupts

* Light Green = HSTX

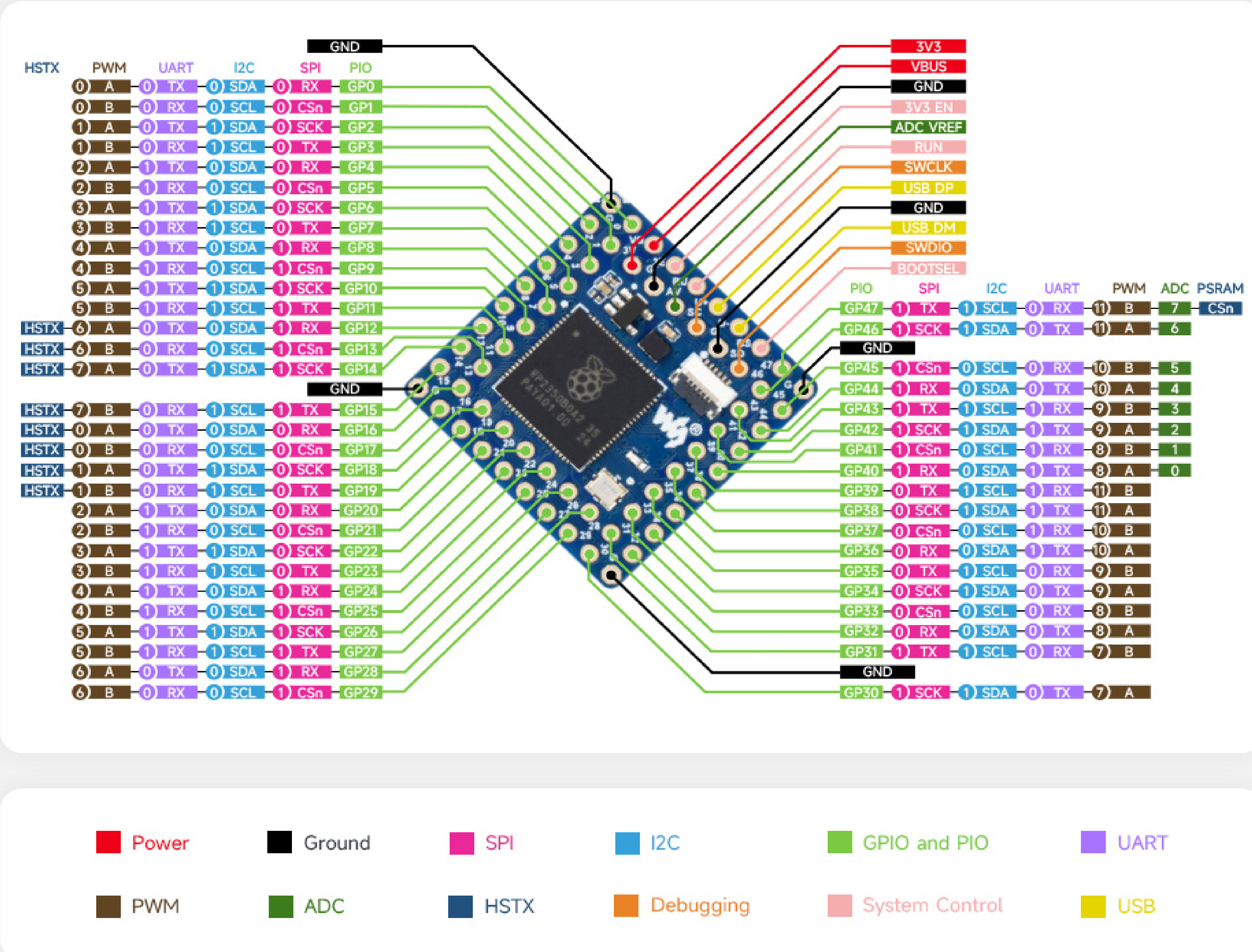
2 × SPI, 2 × I2C, 2 × UART, 6 × 12-bit ADC, 22 × controllable PWM channels, 1 CPU temperature sensor

12 × Programmable I/O (PIO) state machines

Floating point hardware computation

GPIO	ADC	PWM	SPI	I2C	UART
GP24	-------------	PWM 4A	SPI1 RX	I2C0 SDA	UART1 TX
GP25	-------------	PWM 4B	SPI1 CSn	I2C0 SCL	UART1 RX
GP26	-------------	PWM 5A	SPI1 SCK	I2C1 SDA	UART1 CTS
GP27	-------------	PWM 5B	SPI1 TX	I2C1 SCL	UART1 RTS
GP28	-------------	PWM 6A	SPI1 RX	I2C0 SDA	UART0 TX
GP29	-------------	PWM 6B	SPI1 CSn	I2C0 SCL	UART0 RX
GP30	-------------	PWM 7A	SPI1 SCK	I2C1 SDA	UART0 CTS
GP31	-------------	PWM 7B	SPI1 TX	I2C1 SCL	UART0 RTS
GP32	-------------	PWM 8A	SPI0 RX	I2C0 SDA	UART0 TX
GP33	-------------	PWM 8B	SPI0 CSn	I2C0 SCL	UART0 RX
GP34	-------------	PWM 9A	SPI0 SCK	I2C1 SDA	UART0 CTS
GP35	-------------	PWM 9B	SPI0 TX	I2C1 SCL	UART0 RTS
GP43	ADC3	PWM 9B	SPI1 TX	I2C1 SCL	UART1 RTS
GP44	ADC4	PWM 10A	SPI1 RX	I2C0 SDA	UART0 TX
GP45	ADC5	PWM 10B	SPI1 CSn	I2C0 SCL	UART0 RX

PWM	GPIO		
PWM 3B	GP23	LED	
PWM 11A	GP46	ADC6	ADC6 to measure VSYS



GND
HSTX PWM UART I2C SPI PIO
0 A 0 TX 0 SDA 0 RX GP0
0 B 0 RX 0 SCL 0 CSn GP1
1 A 0 TX 1 SDA 0 SCK GP2
1 B 0 RX 1 SCL 0 TX GP3
2 A 1 TX 0 SDA 0 RX GP4
2 B 1 RX 0 SCL 0 CSn GP5
3 A 1 TX 1 SDA 0 SCK GP6
3 B 1 RX 1 SCL 0 TX GP7
4 A 1 TX 0 SDA 1 RX GP8
4 B 1 RX 0 SCL 1 CSn GP9
5 A 1 TX 1 SDA 1 SCK GP10
5 B 1 RX 1 SCL 1 TX GP11
HSTX 6 A 0 TX 0 SDA 1 RX GP12
HSTX 6 B 0 RX 0 SCL 1 CSn GP13
HSTX 7 A 0 TX 1 SDA 1 SCK GP14
GND
HSTX 7 B 0 RX 1 SCL 1 TX GP15
HSTX 0 A 0 TX 0 SDA 0 RX GP16
HSTX 0 B 0 RX 0 SCL 0 CSn GP17
HSTX 1 A 0 TX 1 SDA 0 SCK GP18
HSTX 1 B 0 RX 1 SCL 0 TX GP19
2 A 1 TX 0 SDA 0 RX GP20
2 B 1 RX 0 SCL 0 CSn GP21
3 A 1 TX 1 SDA 0 SCK GP22
3 B 1 RX 1 SCL 0 TX GP23
4 A 1 TX 0 SDA 1 RX GP24
4 B 1 RX 0 SCL 1 CSn GP25
5 A 1 TX 1 SDA 1 SCK GP26
5 B 1 RX 1 SCL 1 TX GP27
6 A 0 TX 0 SDA 1 RX GP28
6 B 0 RX 0 SCL 1 CSn GP29
3V3
VBUS
GND
3V3 EN
ADC VREF
RUN
SWCLK
USB DP
GND
USB DM
SWDIO
BOOTSEL
PIO SPI I2C UART PWM ADC PSRAM
GP47 1 TX 1 SCL 0 RX 11 B 7 CSn
GP46 1 SCK 1 SDA 0 TX 11 A 6
GND
GP45 1 CSn 0 SCL 0 RX 10 B 5
GP44 1 RX 0 SDA 0 TX 10 A 4
GP43 1 TX 1 SCL 1 RX 9 B 3
GP42 1 SCK 1 SDA 1 TX 9 A 2
GP41 1 CSn 0 SCL 1 RX 8 B 1
GP40 1 RX 0 SDA 1 TX 8 A 0
GP39 0 TX 1 SCL 1 RX 11 B
GP38 0 SCK 1 SDA 1 TX 11 A
GP37 0 CSn 0 SCL 1 RX 10 B
GP36 0 RX 0 SDA 1 TX 10 A
GP35 0 TX 1 SCL 0 RX 9 B
GP34 0 SCK 1 SDA 0 TX 9 A
GP33 0 CSn 0 SCL 0 RX 8 B
GP32 0 RX 0 SDA 0 TX 8 A
GP31 1 TX 1 SCL 0 RX 7 B
GND
GP30 1 SCK 1 SDA 0 TX 7 A
Power
Ground
SPI
I2C
GPIO and PIO
UART
PWM
ADC
HSTX
Debugging
System Control
USB

RP2350B

All GPIO pins support level and edge interrupts, Floating Point Hardware Computation

2 × SPI, 2 × I2C, 2 × UART, 8 × 12-bit ADC, 24 × controllable PWM channels, 1 CPU temperature sensor

12 × Programmable I/O (PIO) state machines

Analog to Digital sampling 500 kS/s (using an independent 48 MHz clock)

GPIO	ADC	PWM	SPI	I2C	UART	
GP0	-------------	PWM 0A	SPI0 RX	I2C0 SDA	UART0 TX	
GP1	-------------	PWM 0B	SPI0 Csn	I2C0 SCL	UART0 RX	
GP2	-------------	PWM 1A	SPI0 SCK	I2C1 SDA	UART0 CTS	
GP3	-------------	PWM 1B	SPI0 TX	I2C1 SCL	UART0 RTS	
GP4	-------------	PWM 2A	SPI0 RX	I2C0 SDA	UART1 TX	
GP5	-------------	PWM 2B	SPI0 Csn	I2C0 SCL	UART1 RX	
GP6	-------------	PWM 3A	SPI0 SCK	I2C1 SDA	UART1 CTS	
GP7	-------------	PWM 3B	SPI0 TX	I2C1 SCL	UART1 RTS	
GP8	-------------	PWM 4A	SPI1 RX	I2C0 SDA	UART1 TX	
GP9	-------------	PWM 4B	SPI1 Csn	I2C0 SCL	UART1 RX	
GP10	-------------	PWM 5A	SPI1 SCK	I2C1 SDA	UART1 CTS	
GP11	-------------	PWM 5B	SPI1 TX	I2C1 SCL	UART1 RTS	
GP12	-------------	PWM 6A	SPI1 RX	I2C0 SDA	UART0 TX	HSTX
GP13	-------------	PWM 6B	SPI1 Csn	I2C0 SCL	UART0 RX	HSTX
GP14	-------------	PWM 7A	SPI1 SCK	I2C1 SDA	UART0 CTS	HSTX
GP15	-------------	PWM 7B	SPI1 TX	I2C1 SCL	UART0 RTS	HSTX
GP16	-------------	PWM 0A	SPI0 RX	I2C0 SDA	UART0 TX	HSTX
GP17	-------------	PWM 0B	SPI0 Csn	I2C0 SCL	UART0 RX	HSTX
GP18	-------------	PWM 1A	SPI0 SCK	I2C1 SDA	UART0 CTS	HSTX
GP19	-------------	PWM 1B	SPI0 TX	I2C1 SCL	UART0 RTS	HSTX
GP20	-------------	PWM 2A	SPI0 RX	I2C0 SDA	UART1 TX	
GP21	-------------	PWM 2B	SPI0 Csn	I2C0 SCL	UART1 RX	
GP22	-------------	PWM 3A	SPI0 SCK	I2C1 SDA	UART1 CTS	
GP23	-------------	PWM 3B	SPI0 TX	I2C1 SCL	UART1 RTS	
GP24	-------------	PWM 4A	SPI1 RX	I2C0 SDA	UART1 TX	
GP25	-------------	PWM 4B	SPI1 Csn	I2C0 SCL	UART1 RX	
GP26	*	PWM 5A	SPI1 SCK	I2C1 SDA	UART1 CTS	ADC0
GP27	*	PWM 5B	SPI1 TX	I2C1 SCL	UART1 RTS	ADC1
GP28	*	PWM 6A	SPI1 RX	I2C0 SDA	UART0 TX	ADC2
GP29	*	PWM 6B	SPI1 Csn	I2C0 SCL	UART0 RX	ADC3

RP2350B Below ↓   * RP2040, RP2350A Above ↑

GPIO	ADC	PWM	SPI	I2C	UART
GP30	-------------	PWM 7A	SPI1 SCK	I2C1 SDA	UART0 CTS
GP31	-------------	PWM 7B	SPI1 TX	I2C1 SCL	UART0 RTS
GP32	-------------	PWM 8A	SPI0 RX	I2C0 SDA	UART0 TX
GP33	-------------	PWM 8B	SPI0 Csn	I2C0 SCL	UART0 RX
GP34	-------------	PWM 9A	SPI0 SCK	I2C1 SDA	UART0 CTS
GP35	-------------	PWM 9B	SPI0 TX	I2C1 SCL	UART0 RTS
GP36	-------------	PWM 10A	SPI0 RX	I2C0 SDA	UART1 TX
GP37	-------------	PWM 10B	SPI0 Csn	I2C0 SCL	UART1 RX
GP38	-------------	PWM 11A	SPI0 SCK	I2C1 SDA	UART1 CTS
GP39	-------------	PWM 11B	SPI0 TX	I2C1 SCL	UART1 RTS
GP40	ADC0	PWM 8A	SPI1 RX	I2C0 SDA	UART1 TX
GP41	ADC1	PWM 8B	SPI1 Csn	I2C0 SCL	UART1 RX
GP42	ADC2	PWM 9A	SPI1 SCK	I2C1 SDA	UART1 CTS
GP43	ADC3	PWM 9B	SPI1 TX	I2C1 SCL	UART1 RTS
GP44	ADC4	PWM 10A	SPI1 RX	I2C0 SDA	UART0 TX
GP45	ADC5	PWM 10B	SPI1 Csn	I2C0 SCL	UART0 RX
GP46	ADC6	PWM 11A	SPI1 SCK	I2C1 SDA	UART0 CTS
GP47	ADC7	PWM 11B	SPI1 TX	I2C1 SCL	UART0 RTS

12.4.6. Temperature sensor: ADC8 on RP2350B ADC4 on RP2040 and RP2350A

The on board temperature sensor is very sensitive to errors in reference voltage. At 3.3 V, a value of 891 returned by the ADC corresponds to a temperature of 20.1°C. At a reference voltage 1% lower than 3.3 V, the same reading of 891 correspond to a temperature of 24.3°C: a temperature change of over 4°C. To improve the accuracy of the internal temperature sensor, consider adding an external reference voltage.

$$Temp = 27 - \frac{(ADCvoltage - 0.706)}{0.001721}$$